# SIMATS ENGINEERING
## ASSESSMENT TOOL 2: Case Study

**COURSE NAME: COMPUTER VISION** **COURSE CODE:ITA0518**

**COURSE OUTCOME COVERED**

**CO3:** *Analyze and extract image features using detection and matching techniques for effective image representation and comparison.* (BTL 4)

Assessment Tool Weightage: 20%

Read the given scenario carefully and analyze the problem using appropriate computer vision techniques. Justify your answers with relevant reasoning and comparisons wherever necessary.

**1. Preprocessing and Feature Detection**
A vision system processes images captured under varying lighting conditions. Without preprocessing, feature detection results are inconsistent.
Analyze the issue and explain how preprocessing improves feature detection performance.

ANSWER:

**Analysis of the Problem**
In the given vision system, images are captured under different lighting conditions. Changes in brightness, shadows, and contrast can change the appearance of the same object. As a result, feature detection algorithms may detect different or false keypoints in each image.
Role of Preprocessing
Preprocessing improves the image quality before feature detection. The following techniques can be applied:

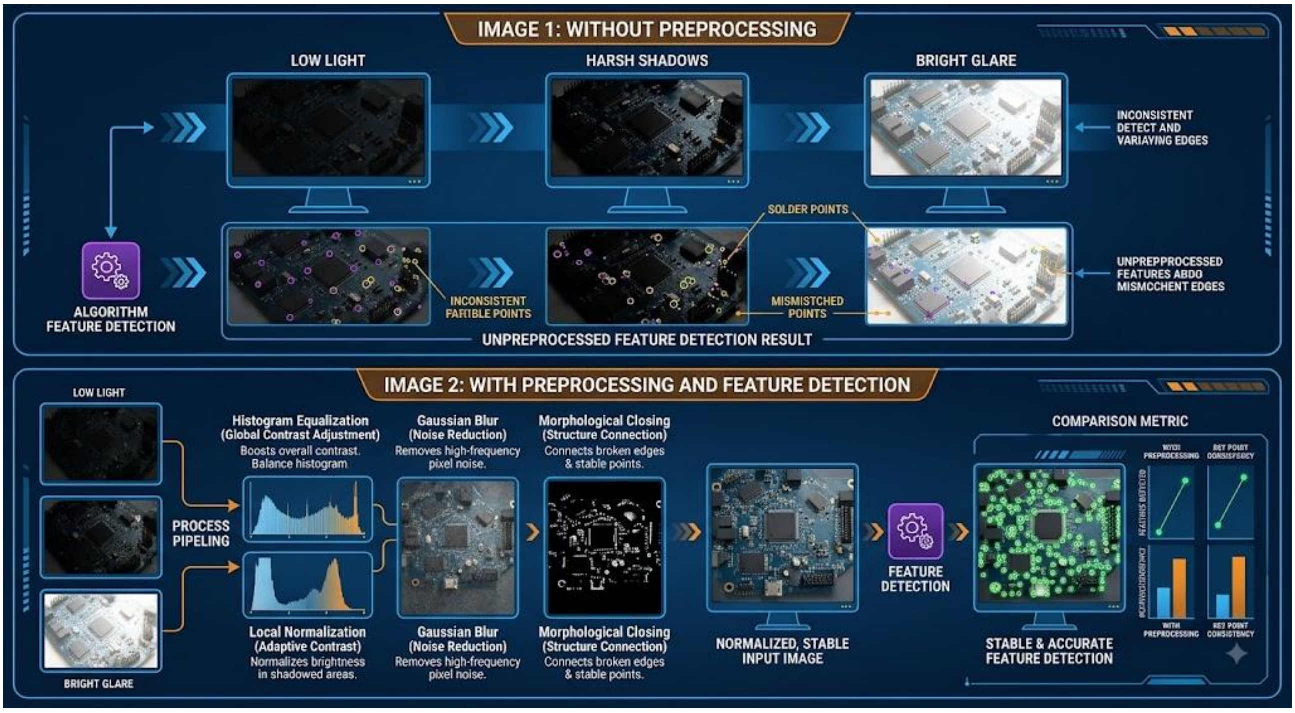
- Grayscale conversion: Reduces unnecessary color information and simplifies processing.
- Noise removal: Gaussian or median filtering removes unwanted noise that may create false features.
- Contrast enhancement: Histogram Equalization or CLAHE improves visibility of edges and corners.
- Illumination normalization: Reduces the effect of uneven brightness and shadows.
- Image sharpening: Enhances important edges and structures.

**Comparison**

| Without Preprocessing | With Preprocessing |
|---|---|
| Features vary with lighting | Features are more consistent |
| More false key points | Fewer false detections |
| Poor edge and corner visibility | Clearer edges and corners |
| Low feature-matching accuracy | Better feature-matching accuracy |
| Less robust system | More robust system |

**Conclusion**

Therefore, preprocessing reduces the effect of lighting variations, noise, and poor contrast, allowing feature detectors such as SIFT, ORB, and Harris Corner Detector to identify more stable and repeatable features. This improves the accuracy, consistency, and robustness of image representation and feature matching.



**4. Feature Matching under Transformations**

Two images of the same object are taken at different scales and orientations. Traditional feature matching fails, but scale invariant feature matching works effectively. Analyze why scale invariant methods perform better in this scenario.

ANSWER:

**Analysis of the Problem**

In the given scenario, two images of the same object are captured at different scales and orientations. Traditional feature matching may fail because the position, size, and direction of the detected features change between the two images.

Why Scale-Invariant Methods Perform Better

Scale-invariant feature methods are designed to detect and describe features even when the image size changes.

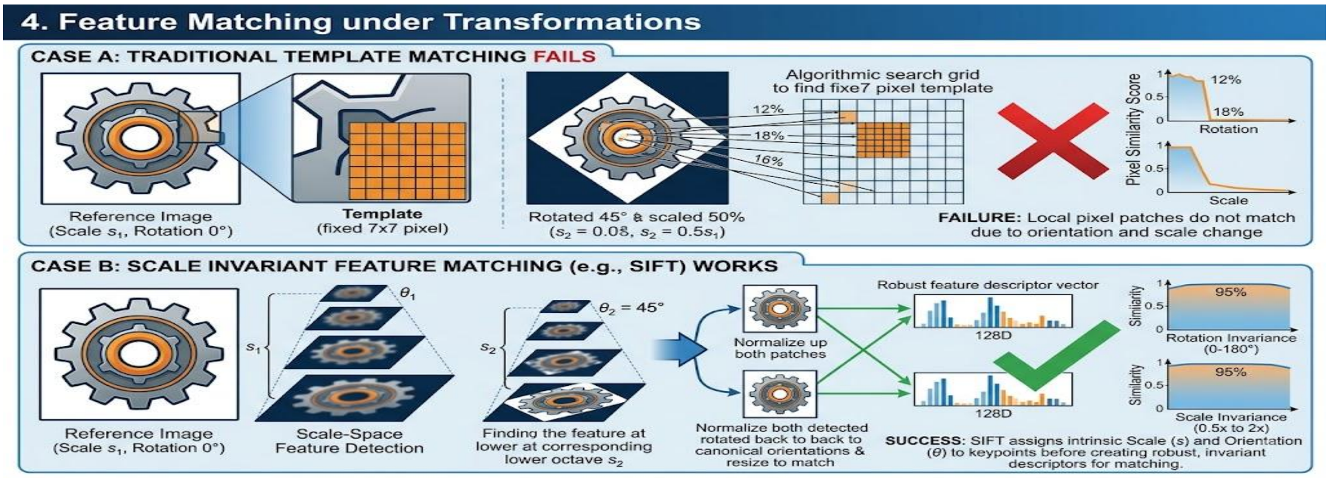
- Scale invariance: The same feature can be recognized even when the object appears larger or smaller.
- Rotation invariance: Features can be matched even when the object is rotated.
- Stable keypoints: Methods such as SIFT detect keypoints at different scales and assign orientations to them.
- Robust descriptors: The feature descriptor represents the local image pattern, allowing corresponding features to be matched despite transformations.

**Comparison**

| Traditional Feature Matching | Scale-Invariant Matching |
|---|---|
| Sensitive to scale changes | Handles scale changes |
| Sensitive to rotation | Handles rotation |
| May produce incorrect matches | Produces more reliable matches |
| Less robust | More robust |

**Conclusion**

Therefore, scale-invariant methods such as SIFT perform better because they are designed to maintain feature consistency under scale and rotation transformations. This results in more accurate and reliable feature matching of the same object.




**7. Corner Detection vs Edge Detection in Tracking**

A tracking system uses edge detection but fails to track moving objects accurately across frames.

Analyze why edges are insufficient and justify the use of corner detection for reliable tracking.

ANSWER:

**Analysis of the Problem**

In the given tracking system, **edge detection alone fails** to track moving objects accurately across frames. Edges represent boundaries between regions, but they usually do not provide enough unique information to identify the exact position of an object.

**Why Edges Are Insufficient**

- Edges can be **long and continuous**, making it difficult to identify a specific point for tracking.
- Edge information may change due to **lighting, noise, or object movement**.
- A moving object may have many similar edges, causing **incorrect matches**.
- Edges do not provide a unique location in both horizontal and vertical directions.

**Why Corner Detection Is Better**

Corners are points where intensity changes significantly in **both horizontal and vertical directions**. They provide distinctive and stable points that can be tracked across consecutive frames.

Methods such as **Harris Corner Detector** and **Shi-Tomasi Corner Detector** can identify these reliable points.

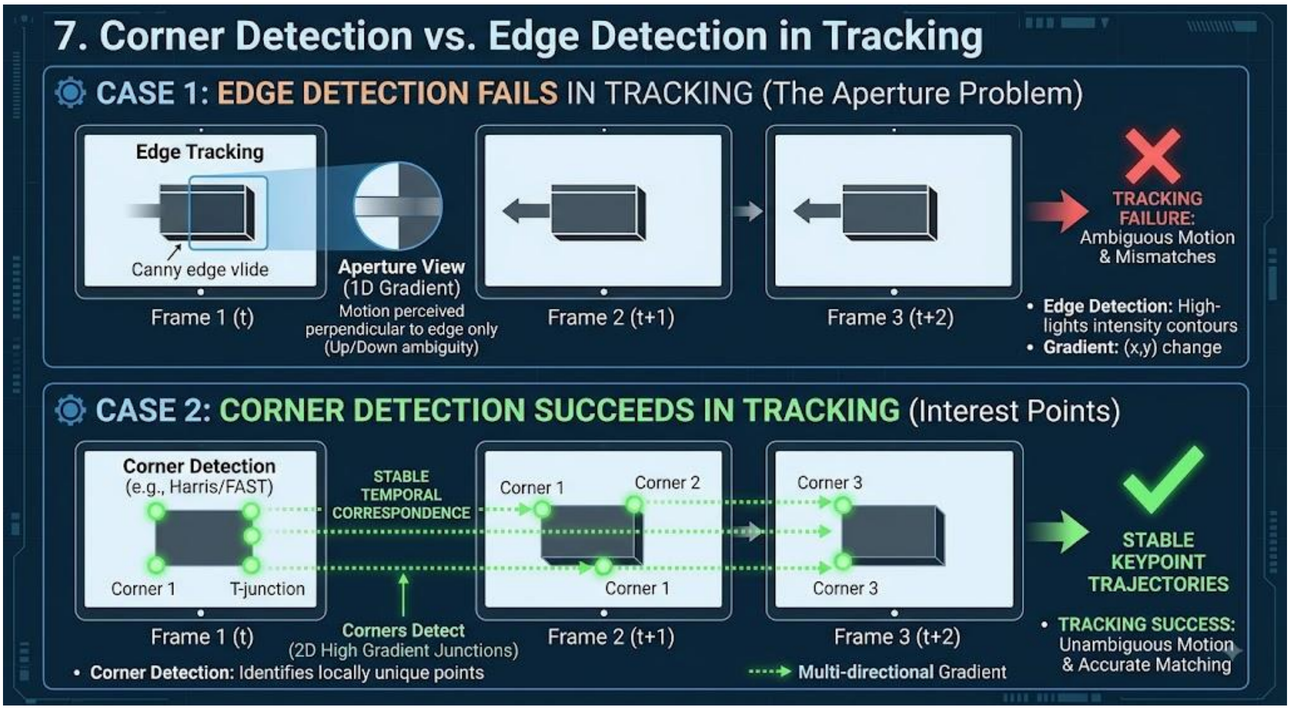
**Comparison**

| **Edge Detection** | **Corner Detection** |
|---|---|
| Detects object boundaries | Detects distinctive points |
| Less unique information | More unique information |
| Difficult to track exact points | Easy to track corresponding points |
| More sensitive to changes | More suitable for point tracking |

| Edge Detection | Corner Detection |
|---|---|
| Less reliable for object tracking | More reliable for object tracking |

**Conclusion**

Therefore, edges alone are insufficient for reliable object tracking because they mainly provide **boundary information**. Corner detection provides **distinctive and localized feature points**, making it more suitable for tracking moving objects accurately across frames.




**8. Hough Transform for Shape Detection**

A system uses Hough Transform to detect circular shapes but fails in complex backgrounds. Analyze the limitations and explain how preprocessing or parameter tuning can improve detection.

ANSWER:

**Analysis of the Problem**

In the given system, the Hough Transform is used to detect circular shapes. However, the presence of a complex background, noise, shadows, and unwanted edges produces many false edge points. These points may be incorrectly interpreted as circles, reducing detection accuracy.

**Limitations**

- Complex backgrounds generate false edges.
- Noise can produce incorrect circle detections.
- Overlapping or partially hidden circles are difficult to detect.
- Incorrect parameter values can cause missed circles or false circles.
- Detecting many possible circle parameters can increase computational cost.

**How Preprocessing Improves Detection**

Before applying the Hough Transform:

- Gaussian/median filtering can reduce noise.
- Grayscale conversion simplifies the image.
- Thresholding or segmentation can separate the object from the background.
- Edge detection such as Canny can provide cleaner edges.
- Morphological operations can remove small unwanted regions.

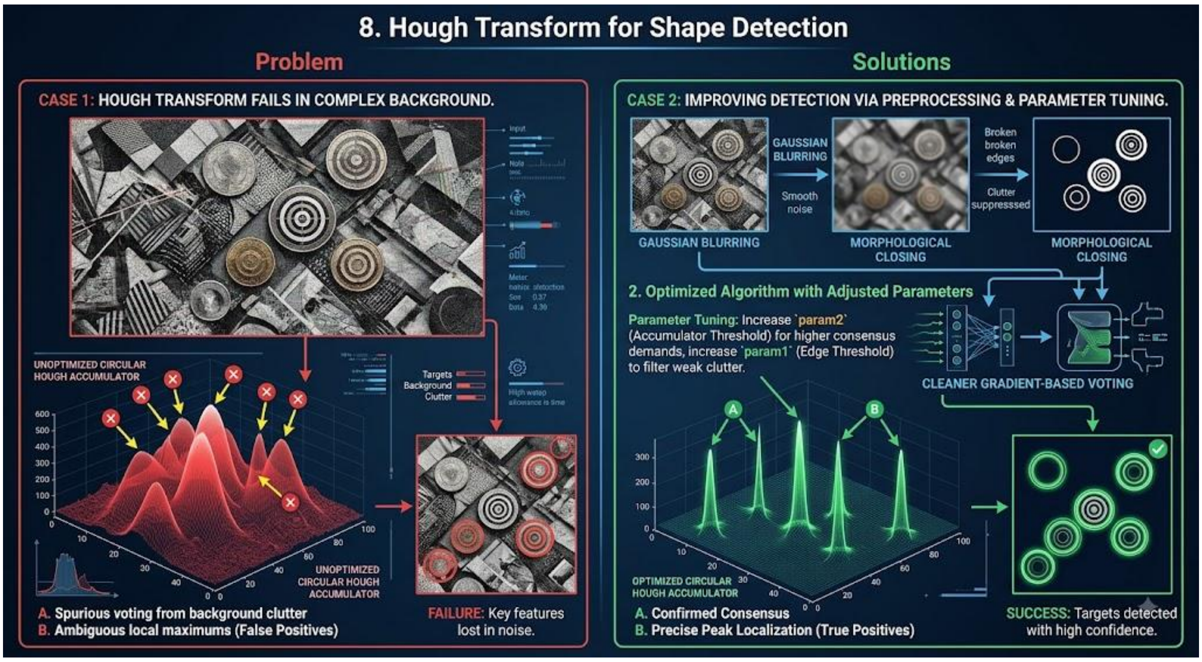
**Parameter Tuning**

The important Hough Circle parameters can be adjusted:

- **Edge threshold:** Controls which edges are considered.
- **Accumulator threshold:** Higher values reduce false detections but may miss weak circles.
- **Minimum distance:** Prevents multiple detections of the same circle.
- **Minimum and maximum radius:** Restricts detection to expected circle sizes.

**Conclusion**

Therefore, Hough Transform performance can be improved by removing noise, reducing background interference, extracting clear edges, and properly tuning the detection parameters. This reduces false positives and improves the accuracy of circular shape detection in complex backgrounds.




**14. PCA and Information Loss**

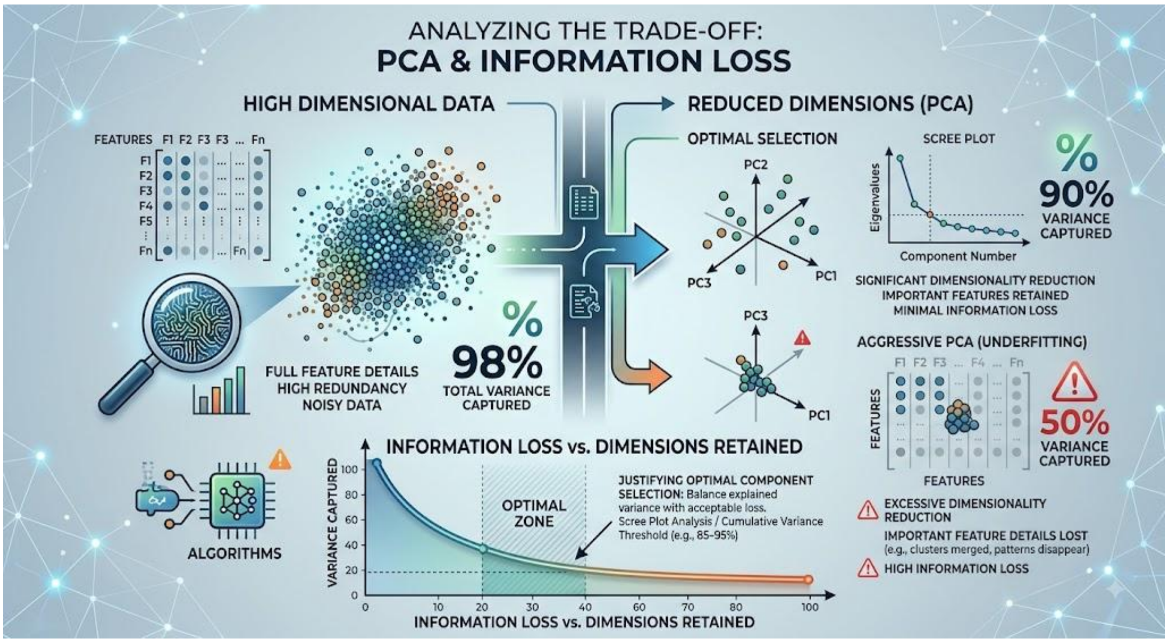
A system applies PCA aggressively and loses important feature details. Analyze the trade-off between dimensionality reduction and information loss, and justify optimal component selection.

ANSWER:

**Answer:**

PCA (Principal Component Analysis) reduces the number of features while preserving the most important information in the data. However, if too many components are removed, important feature details may be lost.

- **Dimensionality Reduction:** PCA transforms the original features into a smaller number of principal components.
- **Information Loss:** Removing components with low variance may also remove useful information, especially when PCA is applied too aggressively.
- **Trade-off:** Fewer components reduce computational cost and complexity but increase information loss. More components preserve information but provide less dimensionality reduction.
- **Optimal Component Selection:** The number of components should be selected based on the explained variance ratio.
- A common approach is to retain components that explain about 90–95% of the total variance.
- A scree plot or cumulative explained variance graph can be used to identify the appropriate number of components.
- The selected components should also be validated using the performance of the final machine-learning model.
- Therefore, the optimal PCA solution balances dimensionality reduction, computational efficiency, and minimum information loss.




***Code of Conduct:***

*I _HARITHI S (192421151)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

HARITHI S

**RUBRICS FOR CALCULATION:**

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Understanding of Case | 25 | Thorough understanding; clearly identifies all key issues | Good understanding with most issues identified | Basic understanding; some issues missed | Poor understanding; problem not identified correctly |
| Application of Concepts | 25 | Effectively applies relevant concepts to analyze the case deeply | Applies appropriate concepts with minor gaps | Limited application of concepts | Fails to apply relevant concepts |
| Analysis | 20 | In-depth analysis with strong reasoning and insights | Good analysis with logical reasoning | Basic analysis with limited depth | Weak or no analysis; lacks reasoning |
| Solution Design | 20 | Innovative, feasible solutions with strong justification | Logical solutions with adequate justification | Basic solutions with weak justification | Incorrect or no solution provided |
| Clarity | 10 | Clear, well-structured, and easy to understand | Organized and understandable presentation | Limited clarity and structure | Poor clarity; difficult to understand |